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Training

Cubesat Modelling Trainee Jun '25-July'25

Aerion Private Limited.

- Conducted Finite Element Analysis (FEA) on CubeSat structures to ensure structural integrity for mission-critical applications like weather monitoring and navigation.
- Modeled satellite components using **Fusion 360** and performed high-fidelity simulations using **Simulia** to analyze mechanical stresses and thermal loads.
- Explored communication subsystem architectures and their integration within the CubeSat frame for surveillance and remote sensing payloads.
- Developed modular 3D designs following standard aerospace prototyping principles to optimize assembly efficiency

Semiconductor Summer Trainee Jun '2-Jun'16

CeNSE, IISc Bangalore

- Gained comprehensive exposure to the end-to-end semiconductor manufacturing flow, spanning from material science to chip packaging and silicon wafer tapeout.
- Analyzed foundry workflows including the transition from RTL to GDSII, ensuring design readiness for physical fabrication.
- Visualized Silvaco TCAD for device-level modeling and process simulation to analyze semiconductor material properties and transistor characteristics.
- Analysed physical layout design and verification using Cadence Virtuoso, focusing on standard cell architectures and interconnect routing

Projects

RRAM-Based In-Memory Computing Spiking Neural Network for Edge AI Feb '26

- Engineered a neuromorphic computing framework utilizing Resistive RAM (RRAM) crossbar arrays to enable energy-efficient in-memory computation for edge AI inference.
- Designed and analyzed 1T1R RRAM cell architecture and implemented crossbar-based synaptic weight storage to perform analog matrix–vector multiplication directly within memory.
- Developed a Spiking Neural Network (SNN) neuron model based on spike-response dynamics to support event-driven computation with ultra-low power consumption.
- Performed Monte Carlo variability analysis to evaluate device-level effects such as threshold variation, conductance fluctuation, and switching instability on spike generation reliability.
- Simulated I–V hysteresis characteristics of RRAM devices and validated resistive switching behavior for synaptic weight modulation.

Tech Stack: Cadence Virtuoso, Verilog/Verilog-A, RRAM Device Modeling, Spiking Neural Networks (SNN), Monte Carlo Analysis,.

Biologically Inspired Synaptic Circuit – Neuromorphic Computing Dec'25

- Designed and implemented a biologically inspired synaptic circuit to emulate neural signal transmission and learning behavior observed in the human brain using CMOS-based neuromorphic architectures.
- Developed a Spiking Neural Network (SNN) model operating at the 90 nm technology node, enabling event-driven computation that mimics neuronal spike generation and synaptic signal propagation.
- Implemented spike-based processing circuits capable of receiving parallel spiking inputs and generating binary output responses, replicating simplified neuronal activation behavior.
- Modeled and simulated neuron and synapse circuits in Cadence Virtuoso, utilizing Cadence Spectre for transient and functional analysis to validate spike timing behavior and circuit stability.
- Analyzed synaptic response characteristics, including spike propagation delay, signal integration, and threshold activation to ensure biologically inspired neural response behavior.

Tech Stack: Cadence Virtuoso & Spectre, Neuromorphic Computing, CMOS Digital/Analog Circuit Design, Spiking Neural Networks

Self-Healing Computing System – Bharat Electronics Limited Sep'25

- Designed and implemented a 4-bit Arithmetic Logic Unit (ALU) with integrated fault detection and Built-In Self-Test (BIST) mechanism to enhance system reliability and enable autonomous fault identification in digital hardware systems.
- Developed a self-healing architecture using partial reconfiguration on a Xilinx FPGA platform, allowing faulty logic blocks to be dynamically isolated and replaced with redundant modules without interrupting system operation.
- Implemented ALU functionalities including arithmetic and logical operations while integrating error detection circuitry to monitor operational correctness during runtime..
- Performed functional verification and hardware testing using simulation and on-board deployment to validate correct operation under multiple input scenarios and fault injection cases

  Tech Stack: Verilog, Xilinx Vivado, FPGA (Nexys4), Digital System Design, Hardware Verification, Fault-Tolerant Computing

Certificates

FPGA Programming and Architecture Oct '25
NIELIT Calicut, NPTEL

Database Management System Oct '25
IIT Kharagpur, NPTEL

RTL to GDS II Nov'25
Digital Badge ,Cadence Design System

Skills

Languages	:	C ++,Verilog.
FPGA	:	NEXY 4 DDR2 ,Spartan 6.
VLSI Skills	:	RTL Design , Verification , Testing , Fabrication .
Hardware & Design	:	Digital Circuits, Analog Circuits, Circuit Design, Neuromorphic Computing, Memory Design.
Simulation & EDA Tools	:	Cadence Spectre, Cadence Virtuoso, Cadence NC-Sim,Xilinx Vivado.
Soft Skills	:	Problem-Solving, Time Management, Team Collaboration, Adaptability.

Achievements

- **Secured First position** Mar '25
  Fit-Fusion, Cognizance, IIT Roorkee - 2025
- **First Runner up** Feb '25
  R2VC workshop, IIT Roorkee - 2025

Education

Lovely Professional University, Punjab **2023 – 2027**
Electronics and communication Engineering — CGPA: 7.77 *Jalandhar, Punjab*

Sri Aurobindo Mira Universal ,school **2021 – 2023**
12th with Science. Percentage-70.8 *Madurai, Tamil Nadu*

Sri Aurobindo Mira Universal , school **2020 – 2021**
10th with Science.Percentage-80.8 *Madurai, Tamil Nadu*